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Economic Growth and Share of Temporary Workers

Nobuyuki Kanazawa

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Abstract A dual labor market structure that comprises "permanent" and "temporary" jobs is common in many Continental European countries and Japan. Moreover, over the last two decades, the share of temporary workers in these countries has increased markedly. In this paper, I construct a search and matching model of a dual labor market with endogenous human capital accumulation and show that, in the presence of two different types of jobs with different rates of wage growth, slowing the aggregate productivity growth rate can prompt a substantial shift in the composition of jobs toward temporary jobs.

Keywords Dual labor market · Growth · Unemployment · Temporary employment
JEL classification E24, J64, O40

1 Introduction

This paper investigates the effect of a permanent reduction in aggregate productivity growth rate on the composition of the labor market when that market comprises two types of employment, referred to as "permanent employment"

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N. Kanazawa
Faculty of Economics, Soka University
1-236 Tangi-machi, Hachio-ji shi, Tokyo, 192-8577
Tel.: +81-42-9502
E-mail: nkanazawa@soka.ac.jp

and "temporary employment."[1] Over the last two decades, the share of temporary workers among all employees in many OECD countries including Continental European countries and Japan has increased markedly. This increasing trend has posed a serious concern for the public in general because temporary workers, on average, earn fewer wages, expect a very low rate of wage growth, receive less generous benefits, and have weaker job security than permanent workers.

This paper studies Japan, which is among the countries that have experienced a significant increase in the share of temporary workers, because: (i) the country experienced both fast growth before 1990 and slow growth after 1990[2]; during the slow growth period, the share of temporary employment nearly doubled from 20 percent to 40 percent; (ii) the presence of two distinctively different types of jobs is well documented[3]; and (iii) panel household survey was readily available, which enabled me to examine different characteristics of permanent and temporary workers.

During the period of slow growth, the share of temporary workers among all employees doubled from 20 percent to nearly 40 percent, as illustrated in 1.[4] The share of temporary workers increases even when the sample is restricted to male employees, particularly male employees at the working age. This outcome implies that the increasing trend is likely to persist even after factors such as greater participation of female workers and aging demographics are eliminated. Notably, the most drastic change is observed among male workers between the ages of 25 and 34. This is attributable to the Japanese employment practice, where the majority of permanent employment contracts are formed when workers are in their 20s and 30s. Therefore, the significant increase in the share of temporary workers for these young men, in contrast to old men whose temporary employment rate increased only modestly, suggests that the job-finding margin contributes more significantly to explaining the increasing share of temporary workers than the job-separation margin does.

In the vast literature on the increasing share of temporary workers, the trend is commonly attributed to changes in labor market regulations that enforce stronger job protection or legislative decisions that facilitate the creation of temporary jobs.[5] While these legislative factors may play a role, this paper argues that, under a certain condition, a slowdown of aggregate productivity growth alone can drive a considerable increase in the share of temporary workers even in the absence of any legislative change. The key mechanism hinges

[1] The referencing of these types of employment in the literature varies. Examples include "good jobs" versus "bad jobs," "standard jobs" versus "non-standard jobs," "full-time work" versus "part-time work," and "indefinite employment" versus "fixed-term employment."

[2] The average Japanese real GDP growth before and after 1990 are 4.47 percent and 0.90 percent, respectively.

[3] Precise definitions of "permanent" and "temporary" jobs are provided in Section 3.

[4] Data are retrieved from the Labor Force Survey, whose design is similar to that of the Current Population Survey in the U.S. The Labor Force Survey was redesigned in 2001. However, an overall increasing trend in the share of temporary workers is evident across the entire population.

[5] For example, Cahuc et al. (2016) and Boeri et al. (2011).

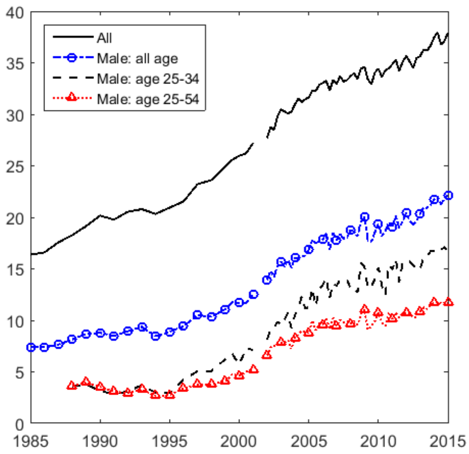

Fig. 1: Proportion of Temporary Workers (percent)

Notes: The figure shows the proportion of temporary workers in Japan; the solid black line denotes all the workers, the blue dotted line denotes male workers, the black dotted line denotes male workers between the ages of 25 and 34, and the red dotted line denotes male workers between the ages of 25 and 54. The data is obtained from "The Special Survey of the Labour Force Survey" from 1984 to 2001 and the "Labour Force Survey (Detailed Tabulation)" since 2002.

on the difference in the rate of human skill accumulation between permanent and temporary workers.

To formalize our intuition, I construct a search and matching model of a dual labor market that is populated by heterogeneous workers, whose skills evolve endogenously according to their employment status. The model incorporates a dual labor market designed to capture two distinctive labor markets for permanent and temporary jobs. Workers are born initially unemployed and with randomly assigned skills. Depending on their skill levels, they decide to seek employment either in the permanent labor market or in the temporary labor market. While choosing the market to search for, unemployed workers face the following trade-off. If the unemployed workers seek employment in the permanent market, they have a chance of finding jobs that enable them to fully utilize their skills and that ensure greater job security and on-the-job training. Therefore, permanent jobs have substantial future benefits to workers, especially faster wage growth. However, the probability of finding such jobs is low. In contrast, if the unemployed workers seek employment in the temporary market, they have a higher probability of finding a job. Nonetheless, workers holding these jobs face weaker job security and cannot utilize their own skills. Instead, they employ a technology that is common to all temporary jobs. Therefore, temporary workers experience only a minimal wage growth consistent with empirical observations. Given the parameter values that I estimate in Section 6, low skilled workers find it optimal to seek temporary jobs while high skilled workers seek permanent jobs; the observation that is consistent with the data.[6]

Given this trade-off between permanent and temporary jobs, I investigate how a slowdown in aggregate productivity growth affects the labor market composition. Suppose an economy enters a slow growth steady state. Slower growth indicates that future income is discounted at a higher rate. This reduces the continuation value of a job because the cost of setting up the initial employment relationship is paid now; in contrast, the benefit of the employment relationship accrues in the future.[7] When the workers' skills are allowed to increase on the job and wages are negotiated during each period, the value of permanent jobs decreases at a significantly faster rate than the rate at which the value of temporary jobs decreases. This is because when the growth rate of aggregate productivity decreases, future wage growth is discounted at a higher rate, making it less valuable to hold jobs that promise higher wage growth (permanent jobs). Faced with the decreasing benefit of permanent jobs, some workers who would seek permanent jobs in a fast-growth economy will seek temporary jobs in a slow-growth economy. Consequently, the share of temporary workers increases in the slow-growth economy. The extent to which the share of temporary workers increases depends on the rate of skill accumulation for the permanent workers. In fact, in Appendix 10.6, I show that when human capital accumulation is muted, slower aggregate productivity growth does not affect the composition of the labor market. When estimated with the Japanese labor market dataset, my model predicts that approximately 38 percent of the observed increase in the share of temporary workers among Japanese males can be explained by its slow growth.

This paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 provides background information about the Japanese labor market and further defines "permanent jobs" and "temporary jobs" in the context of the Japanese labor market. Section 4 illustrates the mechanism using a simple model. Section 5 describes the full model. Section 6 lays out the calibration strategy. Section 7 further shows the results. Section 8 investigates the effects of changing other policy parameters such as firing costs. Section 9 concludes.

2 Related Literature

This paper contributes to the literature on growth and unemployment. Several studies have examined the effect of growth on unemployment in a search and matching framework (Aghion and Howitt (1994), Pissarides (2000), Pissarides and Vallanti (2007)). One of the most heated debates is centered on

[6] See Section 10.2.2.

[7] This effect is often referred to as the "capitalization effect," which is highlighted by Aghion and Howitt (1994) and Pissarides (2000).

the degree to which the new workers embody new technology. If all the technological progress is embodied by new workers, then growth and unemployment are positively correlated. In contrast, if technological growth is disembodied, meaning that both new and existing workers embody technological progress, then growth and unemployment will be negatively correlated. To address this issue, Pissarides and Vallanti (2007) evaluated a model that features embodied and disembodied technology, capitalization, and creative destruction effects. They concluded that embodied technology and creative destruction plays an insignificant role in explaining the steady-state unemployment rate. Following their argument, this paper uses a search and matching model with disembodied technological progress.

The presence of temporary workers in Japan has gained attention in recent years.[8] Asano, Ito and Kawaguchi (2013) empirically examined factors that are most responsible for the rise of temporary employment in Japan and found that changes in the labor force and industrial composition account for only a quarter of the increase in temporary workers. In addition to these factors, this study suggests that the decline of aggregate productivity growth rate and the difference in the rates of human capital accumulation between permanent and temporary jobs are significant sources of the observed rise of the temporary workers share in Japan.

The paper builds on the framework of Huckfeldt (2016), who proposed a search model where the allocation of workers across different labor markets is endogenously determined. He further examines how this allocation changes over the business cycle. In contrast, this study examines how the allocation of workers across distinct jobs changes across growth regimes. In doing so, I identify a separate mechanism by which the impact of changes in growth rates on the allocation of workers depends on the difference in the rates of wage growth across permanent and temporary jobs. This paper is also related to research undertaken by Acemoglu (2001), who developed a search and matching model in which high-wage ("good jobs") and low-wage ("bad jobs") jobs can coexist. Observing the increasing share of temporary employment in Continental European countries, Alonso-Borrego et al. (2005) investigated the role of increased temporary employment in a general equilibrium search and matching model and concluded that temporary jobs increase unemployment, reduce output, and enhance productivity. Caggese and Cuñat (2008) examined the relationship between financial constraints and the labor market composition of permanent and temporary jobs. They found that when firms are likely to be financially constrained, they choose to hire more temporary workers. Examining the experience of labor market reform in France, Blanchard and Landier (2002) discuss the effect of allowing firms to hire workers on fixed-term contracts. They conclude that the main effect of policy reform is a high turnover in entry-level jobs, which results in higher unemployment.

[8] For example, in a structural estimation of the career choices of young workers, Esteban-Pretel, Nakajima and Tanaka (2011) focus on young workers who began their careers as temporary workers. They conclude that in Japan starting a career as a temporary worker has a lasting effect on the welfare of young workers.

In the area of the growth and composition of jobs in the economy, this paper is most closely related to Miyamoto (2016) and Wasmer (1999). In a search and matching framework, Wasmer (1999) showed that through the capitalization effect, the share of temporary workers increases during low technological growth periods. Miyamoto (2016) constructed a search and matching model to quantify the strength of the capitalization effect in Japan. Although our papers are similar, his mechanism crucially depends on the substitutability between goods produced by permanent workers and goods produced by temporary workers. As the rate of substitution between permanent goods and temporary goods decreases, the effects of a change in trend growth on the composition of jobs also decreases. Given the difficulty of measuring the rate of substitution between goods produced by permanent workers and goods produced by temporary workers in the real world, I propose an alternative mechanism that relies on the difference in the rate of wage growth between the two types of employment by integrating heterogeneous workers' skills that evolve dynamically.

3 "Permanent" and "Temporary" Jobs in the Japanese Labor Market

This section outlines the institutional background of the Japanese labor market, whose composition is the focus of this paper. Particularly, I describe standard Japanese labor market practices that many believe differ from US practices, and further define "permanent employment" and "temporary employment" in the context of Japanese labor market practices. Kambayashi and Kato (2012) summarize a set of standard practices that are coherently observed in the Japanese labor market as follows:

1. Strong employment protection against lay-offs, which results in long job tenure.
2. Employee involvement in problem-solving activities from the bottom-up, which encourages workers to exert discretionary effort and gain local knowledge.
3. Careful screening and extensive on-the-job training that increases worker ability[9]

This paper refers workers who are covered by these standard practices to as "permanent workers." Another group of Japanese workers has an employment relationship that does not conform to these practices. They constitute the second segment of the Japanese labor market, a situation I refer to as "temporary employment." These workers are often paid lower wages, receive

[9] Kambayashi and Kato (2012) also highlighted two other Japanese employment practices: (4) A robust scheme to share information through unions so that information asymmetry between employees and between employees and management is avoided; and (5) Incentive schemes that include employee ownership of the firms and profit-sharing, which encourage workers and the firm to develop a sense of shared interests.

less generous benefits, enjoy less control over their work, and have weaker employment protection than permanent workers (Kambayashi and Kato (2012)). In this regard, the permanent and temporary jobs of the Japanese labor market are similar to the "good jobs" and "bad jobs" that are described in a good job and bad job literature and "open-ended jobs" and "fixed-term jobs" in the literature that examines European labor markets [10].

Appendix 10.2, documents three empirical facts about permanent and temporary workers that are relevant for the model proposed in the current paper: (i) the rate of wage growth is higher for permanent workers, (ii) people with high school or middle school degrees are more likely to become temporary workers while people with college or higher degrees are more likely to become permanent workers, and (iii) the opportunity to receive training on the job for temporary workers is approximately half of that for permanent workers. These facts highlight the importance of skills and skill accumulation when workers decide the types of jobs they seek. The following section presents the model.

4 Simple Model

For an illustration of a mechanism, I first construct a simple model that features two types of jobs. For convenience, I refer to the first type of job as a permanent job and the second type of job as a temporary job. These two types of jobs are identical except for the fact that wages for permanent workers exhibit growth while wages for temporary workers remain fixed for their entire career. Workers employed in the permanent jobs initially receive wages, w_{perm}. In the next period, their wages increase to w'_{perm}, which will then be fixed for the rest of their careers. I further assume that once attached, the employed workers do not separate from the jobs. Besides the wage growth that is specific to the permanent workers, the wages of both sectors will increase at the common rate of γ_X, which reflects the growth rate of the aggregate productivity.

Let W_{perm} denote the present value of a permanent worker. Then,

$$W_{\text{perm}} = w_{\text{perm}} + \beta\gamma_X \frac{w'_{\text{perm}}}{1-\beta\gamma_X}, \tag{1}$$

where β is a discount factor.

Temporary workers do not experience wage growth. Therefore, the value of a temporary worker is:

$$W_{\text{temp}} = \frac{w_{\text{temp}}}{1-\beta\gamma_X}, \tag{2}$$

An unemployed worker is indifferent between searching for a permanent job and a temporary job if the following condition holds:

$$W_{\text{perm}} = W_{\text{temp}},$$

[10] See, for example, Acemoglu (2001) and Kalleberg (2011)

This condition requires the following wage structure: $w'_{\text{perm}} > w_{\text{temp}} > w_{\text{perm}}$.

Suppose that the technological growth that is common to both types of workers decreases (γ_X decreases). Then, the values of both permanent workers and temporary workers decrease. For the unemployed workers to continue being indifferent between choosing between permanent jobs and temporary jobs, the values of a permanent worker and that of a temporary worker must decrease at the same rate, which implies $dW_{\text{perm}}/d\gamma_X = dW_{\text{temp}}/d\gamma_X$. In contrast, the unemployed workers prefer temporary jobs if the value of permanent workers decreases at a greater rate than the rate at which the value of temporary workers decreases: $dW_{\text{perm}}/d\gamma_X > dW_{\text{temp}}/d\gamma_X$.

Apparently, if the permanent jobs feature wage growth while the temporary jobs do not, then the rate of reduction in the value of permanent jobs is larger in response to the decrease in γ_X as follows:

$$\frac{dW_{\text{perm}}}{d\gamma_X} = \frac{\beta w'_{\text{perm}}}{(1-\beta\gamma_X)^2} > \frac{\beta w_{\text{temp}}}{(1-\beta\gamma_X)^2} = \frac{dW_{\text{temp}}}{d\gamma_X},$$

since $w'_{\text{perm}} > w_{\text{temp}}$.

Because changes in the discount factor have effects only through wages in the future periods, the rate of response is larger for the permanent sector where the future wage is higher than the temporary wages. Therefore, the unemployed workers who were previously indifferent about permanent jobs and temporary jobs now prefer to be employed in temporary jobs when γ_X decreases.

To fully evaluate the extent to which slow growth affects the labor composition, I construct a full model in the next section. Constructing a search and matching model, as opposed to the simple model illustrated in this section, has the following advantages. First, the simple model predicts that the wages in the initial stage are lower for all permanent workers than temporary workers. A fully-specified search and matching model yields more realistic wage dynamics wherein observed wages are almost always higher for permanent workers.[11] Second, the search and matching model is useful for a quantitative assessment of the mechanism. By constructing an otherwise standard search and matching model that is often used to characterize a labor market, I assess the quantitative significance of the proposed mechanism through which the share of temporary workers increases among Japanese male employees.[12] Similarly, the third advantage of the search-matching model is that it is useful for assessing the quantitative importance of other potential factors that

[11] However, the full model predicts that a small fraction of permanent workers would still accept lower permanent wages in the initial periods of employment if the difference between initial permanent wages and temporary wages are sufficiently small.

[12] The fully-specified search and matching model considers the adjustments in labor market tightness and wages as a result of a slowdown in the aggregate productivity growth. For example, when more people seek temporary jobs, the temporary labor market becomes more congested, which makes finding a temporary job more difficult and searching in the temporary labor market less attractive ("congestion effects"), attenuating the initial effect of the slowdown in technological growth on the job composition.

could contribute to the increasing share of temporary workers. In Section 8, I investigate the importance of changing firing cost of permanent workers and improving temporary job technology on the labor market composition.

5 Full Model

5.1 Environment

The model builds on Huckfeldt (2016) who incorporates elements of the Diamond-Mortensen-Pissarides search and matching model and the Ljungqvist and Sargent (1998) model of human capital accumulation and depreciation. The economy is populated by risk-neutral workers and firms. Each employment relationship comprises a worker-firm pair. Workers are born unemployed and have skills that are drawn from a log-normal distribution: $\log(h) \sim N(0, \sigma_h^2)$. The skills of each worker dynamically evolve according to their employment status.

Two labor markets comprise my model. The first is the permanent job market and the second is the temporary job market. These two employment types differ in two respects: (i) while a firm must pay a firing cost to dissolve a permanent match, there is no cost of dissolving a temporary match; and (ii) permanent workers use their skills in the production process, in contrast to temporary workers, who do not utilize their skills. Instead, temporary workers use a technology that is commonly available to all temporary workers. I assume that firing costs constitute waste rather than transfer to workers (i.e., administrative and legal costs). Once an employment relationship is formed, firms produce goods according to the following linear technologies:

$$\tilde{y}_{\text{perm}} = \underbrace{X}_{\substack{\text{labor-augumenting}\\ \text{technology}}} \times \underbrace{h}_{\text{skill level}}, \qquad \tilde{y}_{\text{temp}} = \underbrace{X}_{\substack{\text{labor-augumenting}\\ \text{technology}}} \times \underbrace{\bar{A}}_{\substack{\text{temporary}\\ \text{technology}}}$$

The productivity of temporary workers is fixed at $\bar{A}$. To analyze the effect of growth, I assume that there is a labor-augmenting technology, X.[13] Additionally, I assume that all separations occur for exogenous reasons because the fluctuation in Japanese job separation rate is limited. That there is no on-the-job search reflects the fact that in Japan, a direct transition from temporary employment to permanent employment is rare. Esteban-Pretel et al. (2011), who conducted a structural estimation of the career choices of young workers in Japan, found that temporary employment is seldom a stepping stone to permanent employment. Finally, I focus on stationary stochastic equilibrium.

5.2 Construction of Labor Markets

There are two labor markets. During each period, unemployed workers with skill level, h, choose the market to seek a job. The search is random within

[13] See Section 5.3 and 10.3 for more detail

the same market. The match is formed according to a Cobb-Douglas function that considers job-seekers and vacancies factors of input. Let M denote the number of matches and let u and v be the number of job-seekers and vacancies, respectively. The match is formed according to the following:

$$M_j = \overline{m}_j u_j v_j^{1-\psi}, \quad \text{where } j \in \{\text{perm, temp}\},$$

where $\overline{m}_j$ is a constant match efficiency for j labor market. The match efficiency is different across two markets. For convenience, I define market tightness as the ratio of vacancies per job-seeker: $\theta_j = v_j/u_j$. Further, the job-finding probability can be defined as $f(\theta_\mathrm{j}) = \bar{m}_\mathrm{j}\theta_\mathrm{j}^{1-\psi}$ and the job-filling probability as $q(\theta_\mathrm{j}) = \bar{m}_\mathrm{j}\theta_\mathrm{j}^{-\psi}$ for $j \in \{\text{perm}, \text{temp}\}$.

The presence of the firing cost changes the value functions of the permanent workers and firms as well as the wage structure of the permanent workers. Following Mortensen and Pissarides (1999), Pries and Rogerson (2005) and Pissarides (2000), I assume that firms that reject new permanent workers do not have to pay firing costs because their employment relationships have not yet started. Once they start working, the firms must pay firing costs when their matches are destroyed. This setting gives rise to a two-tier wage structure for permanent jobs, which is often described as an "inside" wage and "outside" wage (Lindbeck et al. (1989)).

w^N_{perm} denotes the wage for a newly employed permanent worker and w^E_{perm} denotes the wage of an existing permanent worker. Because of the presence of the two-tier wage structure, there have to be two value functions for the permanent workers and jobs.

5.3 Balanced Growth

Following Pissarides (2000) and Pissarides and Vallanti (2007), my model assumes disembodied technological growth. I assume that there is a labor-augmenting technology, X_t that exhibits a steady-state rate of growth as $\gamma_X = X_{t+1}/X_t$ for all t. Furthermore, I assume that all exogenous variables grow at the constant rate of γ_X. Therefore, the cost of vacancy, the unemployment income, and the firing cost all grow at the rate of γ_X. Under this assumption, the economy is on a balanced growth path. In the following sections, I present a model under which all variables are divided by X (i.e. $y_{\text{perm}} = \tilde{y}_{\text{perm}}/X$). For a detailed derivation, please refer to Section 10.3.

5.4 Human skill dynamics

The workers in my model are endowed with skills. The set of skills are measured between $\underline{h}$ and $\overline{h}$, and each skill level is located at Δ_h distance from the others. Human capital in my model evolves dynamically. Given that wages are negotiated during each period, the wages grow as workers' skills upgrade while on the job. According to Rubinstein and Weiss (2006), there are two main sources for an observed earnings/experience profile in the U.S.: (1) skill accumulation and (2) job seeking (including outside job offers). In my model, workers increase their wages only by accumulating more skills. My model ignores the wage growth from an on-the-job search because in Japan, the job-to-job transition rate has historically been small.

As noted in Section 10.2.1, the rate of wage growth differs across the two types of employment. To capture the different rates of wage growth, π_{perm} denotes the probability of a skill upgrade by Δ_H for permanent workers and π_{temp} denotes the probability of a skill upgrade for temporary workers. Therefore, during each period, the π_j fraction of workers in sector j who have an h skill-level upgrade their skills to $h+\Delta_h$ in the next period. In this model, however, although temporary workers can upgrade their skills, they cannot utilize their skills at their jobs. Additionally, unemployed workers face the probability of skill deterioration with the probability, π_{u}. Therefore, for unemployed workers who have skill-level, h, their skills go down to $h-\Delta_h$ during each period with the probability π_u.

5.5 Value of Unemployment

Let $U(h)$ denote the present value of an unemployed worker who has h skill-level.

$$U(h) = b + \beta(1-\nu)\gamma_X \max \mathbb{E}\left[U_{\text{perm}}(h'), U_{\text{temp}}(h')\right], \tag{3}$$

where ν is the exogenous probability of retirement, and β is the discount factor. Unemployed workers first receive an unemployment income, b, and decide whether to seek employment in the permanent or temporary markets.

Let $U_{\text{perm}}(h)$ denote the present value of unemployed workers who seek employment in the permanent labor market, and let $U_{\text{temp}}(h)$ the present value of unemployed workers who seek employment in the temporary market. Then,

$$U_{\text{perm}}(h) = f(\theta_{\text{perm}})(W^N_{\text{perm}}(h) - U(h)) + U(h), \tag{4}$$

$$U_{\text{temp}}(h) = f(\theta_{\text{temp}})(W_{\text{temp}}(h) - U(h)) + U(h), \tag{5}$$

where $W^N_{\text{perm}}(h)$ and $W_{\text{temp}}(h)$ are the present values of a newly hired permanent worker and a temporary worker, respectively. Once the unemployed worker decides the market from which to seek employment, the unemployed worker finds a job with the probability $f(\theta_{\text{j}})$ for $j \in \{\text{perm}, \text{temp}\}$. If the unemployed worker could not find a job, then they will remain unemployed when the next period starts.

For reasonable parameter values, all workers with a skill level above a given cutoff h^* choose to seek employment in the permanent market because the value of seeking employment in the permanent market exceeds the value of seeking employment in the temporary market. Therefore, for $h > h^*$, $U_{\text{perm}}(h) > U_{\text{temp}}(h)$. Similarly, all workers below cutoff h^* find it optimal to seek employment in the temporary market. Therefore, an unemployed worker who held a temporary job previously might find it optimal to seek a permanent job if he accumulated sufficient skills when he worked the temporary job. In contrast, an unemployed worker who previously held a permanent job will seek employment in the temporary market if his skill drops to a level below h^*.

5.6 Value of Worker

Let $W^N_{\text{perm}}(h)$ denote the present value of a new permanent worker with skill level h:

$$W^N_{\text{perm}}(h) = w^N_{\text{perm}}(h) + \beta(1-\nu)\gamma_X \, \mathbb{E}\left[(1-\delta_{\text{perm}})(W^E_{\text{perm}}(h') - U(h')) + U(h')\right], \tag{6}$$

subject to the law of motion for h. $w^N_{\text{perm}}(h)$ denotes the wage of a new permanent worker with skill level h. An existing permanent worker faces an exogenous probability of match destruction, δ_{perm}, at the end of each period. If the match is not destroyed, then the worker stays in the same firm.

Let $W^E_{\text{perm}}(h)$ denote the present value of an existing permanent worker:

$$W^E_{\text{perm}}(h) = w^E_{\text{perm}}(h) + \beta(1-\nu)\gamma_X \, \mathbb{E}\left[(1-\delta_{\text{perm}})(W^E_{\text{perm}}(h') - U(h')) + U(h')\right], \tag{7}$$

subject to the law of motion for h. $w^E_{\text{perm}}(h)$ denotes the wage of an existing permanent worker with skill level h. The only difference between the value of a new worker and that of an existing worker is the wage.

Finally, let $W_{\text{temp}}(h)$ denote the present value of a temporary worker:

$$W_{\text{temp}}(h) = w_{\text{temp}}(h) + \beta(1-\nu)\gamma_X \, \mathbb{E}\left[(1-\delta_{\text{temp}})(W_{\text{temp}}(h') - U(h')) + U(h')\right], \tag{8}$$

subject to the law of motion for h. $w_{\text{temp}}(h)$ denotes the wage of a temporary worker with skill level h, and δ_{perm} denotes the exogenous probability of match destruction for a temporary worker.

5.7 Value of a Job

Let $J^N_{\text{perm}}(h)$ denote the present value of a new permanent job

$$\begin{aligned} J^N_{\text{perm}}(h) &= y_{\text{perm}} - w^N_{\text{perm}}(h) \\ &\quad + \beta(1-\nu)\gamma_X \, \mathbb{E}\left[(1-\delta_{\text{perm}})(J^E_{\text{perm}}(h') - V) + V - \delta_{\text{perm}}\phi\right], \end{aligned} \tag{9}$$

subject to the law of motion for h. V is the value of posting a vacancy. The match continues to the next period unless it is not exogenously destructed with the probability δ_{perm}. If the match is destroyed, the firms pay the firing costs of ϕ at the end of the period.

Let $J^E_{\text{perm}}(h)$ denote the present value of an existing permanent job

$$\begin{aligned} J^E_{\text{perm}}(h) &= y_{\text{perm}} - w^E_{\text{perm}}(h) \\ &\quad + \beta(1-\nu)\gamma_X\, \mathbb{E}\left[(1-\delta_{\text{perm}})(J^E_{\text{perm}}(h') - V) + V - \delta_{\text{perm}}\phi\right], \end{aligned} \tag{10}$$

subject to the law of motion for h. The value of an existing permanent job and the value of a new permanent job differ only in their wages. As explained above, wages differ because new permanent firms do not need to pay firing costs when they reject new workers.

Similarly, let $J_{\text{temp}}(h)$ define the present value of a temporary job:

$$\begin{aligned} J_{\text{temp}}(h) &= y_{\text{temp}} - w_{\text{temp}}(h) \\ &\quad + \beta(1-\nu)\gamma_X\, \mathbb{E}\left[(1-\delta_{\text{temp}})(J_{\text{temp}}(h') - V) + V\right], \end{aligned} \tag{11}$$

subject to the law of motion for h. The value of temporary jobs is similar to that of permanent jobs except that temporary jobs do not incur firing costs.

5.8 Value of Vacancy and the Free Entry Condition

Finally, the value of posting a vacancy in the permanent market is

$$V_{\text{perm}} = -\kappa_{\text{perm}} + \beta(1-\nu)\gamma_X\, \mathbb{E}\left[q(\theta_{\text{perm}}) \int_h (J^N_{\text{perm}}(h) - V_{\text{perm}}) d\mu^h_{\text{perm}} + V_{\text{perm}}\right], \tag{12}$$

where μ^h_{perm} is the skill distribution of unemployed workers in the permanent market. I assume that at the time of posting vacancies, the firms do not know the skills of the job-seekers. The worker's skill is revealed only after he or she meets the firm. Thus, the job-filling rate is the same across all skill-levels. The cost of posting a vacancy is defined as κ_{perm} for permanent jobs. The worker and firm meet at a job-filling probability given by $q(\theta_{\text{perm}})$. If the vacancy meets a worker, then a new job whose continuation value is given by $J^N_{\text{perm}}(h)$ is formed.

The value of posting a vacancy in the temporary market is

$$V_{\text{temp}} = -\kappa_{\text{temp}} + \beta(1-\nu)\gamma_X\, \mathbb{E}\left[q(\theta_{\text{temp}}) \int_h (J_{\text{temp}}(h) - V_{\text{temp}}) d\mu^h_{\text{temp}} + V_{\text{temp}}\right], \tag{13}$$

where μ^h_{temp} is the skill distribution of unemployed workers in the temporary market and κ_{temp} is the cost of posting a vacancy in the temporary market. Each vacancy meets the job-seeker with the probability of $q(\theta_{\text{temp}})$. In equilibrium, the free-entry condition drives the value of a vacancy in both markets down to zero: $V_{\text{perm}} = 0$ and $V_{\text{temp}} = 0$.

5.9 Wage Determination

Following Pissarides (2000), I assume that workers and firms negotiate wages such that they split the matching surplus according to the Nash Bargaining game. Let η denote a worker's fixed bargaining power. According to the Nash Bargaining solution, the new permanent worker and the firm split the match surplus as follows:

$$(1-\eta)(W^N_{\text{perm}}(h) - U_{\text{perm}}(h)) = \eta J^N_{\text{perm}}(h), \tag{14}$$

The existing permanent worker and a firm split the match surplus as follows:

$$(1-\eta)(W^E_{\text{perm}}(h) - U_{\text{perm}}(h)) = \eta (J^E_{\text{perm}}(h) + \phi), \tag{15}$$

Finally, for the temporary worker and firm, the match surplus is divided as follows:

$$(1-\eta)(W_{\text{temp}}(h) - U_{\text{temp}}(h)) = \eta J_{\text{temp}}(h), \tag{16}$$

Then, wages for new permanent workers can be shown as follows:[14]

$$\begin{aligned}
w^N_{\text{perm}} &= (1-\eta)b + \eta y_{\text{perm}} - \eta\beta(1-\nu)\gamma_X \phi \qquad (17)\\
&\quad + (1-\eta)\beta(1-\nu)\gamma_X \left\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h-\Delta_h) - \pi_{\text{perm}} U(h+\Delta_h) \right\}\\
&\quad + (1-\eta)\beta(1-\nu)\gamma_X \left\{ (1-\pi_u) f(\theta_{\text{perm}}) \left(W^N_{\text{perm}}(h) - U(h) \right) \right.\\
&\qquad\qquad \left. + \pi_u f(\theta_{\text{j}}) \left(W^N_{\text{j}}(h-\Delta_h) - U(h-\Delta_h) \right) \right\},
\end{aligned}$$

The basic intuition of the Nash Bargaining solution holds in this wage equation 17. The wage is the weighted average of the workers' outside options and firm profits. A worker's outside option is the unemployment income, b, plus the expected value of finding another job if the worker becomes unemployed. Note that there is a possibility that the workers' skills will drop to a level below h^* during the next period of unemployment. If their skills drop to a level below h^*, they will search in the temporary market. Therefore, j in the last line of equation 17 could be either “perm” or “temp,” depending on the workers' skill level. Firms and workers also consider the value of becoming unemployed in the next period in case of a skill upgrade and in the case of a skill deterioration, both of which are captured in the equation's second line (17). Additionally, new workers are willing to accept lower initial wages if they can start working at firms and become existing workers during next period. After they become existing workers, they can use the firing costs as a credible threat against the firm to demand a higher wage.

[14] See Appendix 10.5 for the derivation.

For existing permanent workers, the wages are given by:

$$
\begin{aligned}
w^E_{\text{perm}} = {} & (1-\eta)b + \eta y_{\text{perm}} + \eta(1-\beta(1-\nu)\gamma_X)\phi \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Bigg\{ (\pi_{\text{perm}} - \pi_u)U(h) + \pi_u U(h-\Delta_h) - \pi_{\text{perm}} U(h+\Delta_h) \Bigg\} \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Bigg\{ (1-\pi_u) f(\theta_{\text{perm}}) \Bigg(W^N_{\text{perm}}(h) - U(h) \Bigg) \\
& \qquad + \pi_u f(\theta_{\text{j}}) \Bigg(W^N_{\text{j}}(h-\Delta_h) - U(h-\Delta_h) \Bigg) \Bigg\},
\end{aligned} \tag{18}
$$

Note that because existing permanent workers know that their employers have to pay firing costs if the matches dissolve, the workers can demand higher wages. In our wage equation for the existing permanent worker, this appears in the form of $\eta\phi$ in the first line of equation 18, which gives the additional wage premium for the existing permanent worker.

For temporary workers, the wages are as follows:

$$
\begin{aligned}
w_{\text{temp}} = {} & (1-\eta)b + \eta y_{\text{temp}} \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Bigg\{ (\pi_{\text{temp}} - \pi_u)U(h) + \pi_u U(h-\Delta_h) - \pi_{\text{temp}} U(h+\Delta_h) \Bigg\} \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Bigg\{ (1-\pi_u) f(\theta_{\text{temp}}) \Bigg(W_{\text{temp}}(h) - U(h) \Bigg) \\
& \qquad + \pi_u f(\theta_{\text{temp}}) \Bigg(W_{\text{temp}}(h-\Delta_h) - U(h-\Delta_h) \Bigg) \Bigg\},
\end{aligned} \tag{19}
$$

Because no firing cost is incurred for temporary jobs, ϕ does not enter the wage equation for temporary workers.

5.10 Dynamics of Distribution

Finally, the unemployment rate for each skill-level evolves as follows:

$$
\begin{aligned}
u'(h) = {} & (1-\pi_u)(1-f(\theta_j)) \cdot u(h) + \pi_u(1-f(\theta_j)) \cdot u(h+\Delta_h) \\
& + (1-\pi_{\text{j}})\delta_{\text{j}} \cdot (1-u(h)) + \pi_{\text{j}}\delta_{\text{j}} \cdot (1-u(h-\Delta_h)),
\end{aligned} \tag{20}
$$

where j in equation 20 could be “perm” or “temp” depending on the skill level, h.

Table 1: Externally calibrated parameters

Variable	Description	Value / Source
R	Interest Rate	$1.0485^{1/12}$, average real interest rate between 1981-1990
β	Discount rate	$1/R$
δ_{perm}	Separation rate for perm workers	0.0035
δ_{perm}	Separation rate for temp workers	0.0071
γ_X	Technological growth rate	$1.0413^{1/12}$, average aggregate growth labor productivity rate between 1981-1990
θ_{perm}	Labor market tightness in perm sector	1.0, normalization
θ_{temp}	Labor market tightness in temp sector	2.24, Miyamoto (2016)
ψ	Matching elasticity	0.6, Miyamoto (2011)
η	Nash bargaining power for workers	0.6, see text
ν	Retirement probability	0.00208, see text
$\overline{h}$	max skill level	5
$\underline{h}$	min skill level	0
Δ_h	Human capital increment	0.0168, 300 equispaced grid
σ_h	std. of initial skill dist.	0.494, see text

Notes: Data for real interest rates is collected from the World Bank data. Data for aggregate labor productivity, defined as the real GDP over hours worked, is collected from the OECD statistics.

6 Calibration

The model is calibrated to match data at the monthly frequency. Table 1 summarizes the externally calibrated parameters.

I set the interest rate $R = 1.0485^{1/12}$, which is the average short-term real rate between 1981 and 1990. For the discount factor, I simply take the inverse of the interest rate. I set the steady-state rate of technological growth, $\gamma_X = 1.0413^{1/12}$, which translates into an average annual growth rate of aggregate labor productivity, defined as the real GDP over hours worked, between 1981 and 1990.

I calibrate the average monthly separation rate for permanent workers to be 0.0035, which implies that their average years of working for a firm is 14.95 years. Following Miyamoto (2016), I calculate that the ratio of the separation rate for permanent workers to temporary workers is 0.49, as indicated in the Survey on Employment Trends conducted by the Ministry of Health, Labor and Welfare. I assume that matching efficiency parameters, $\bar{m}_j$, differ across sectors but that the elasticity parameters, ψ, are the same across sectors. Lin and Miyamoto (2012) estimated that the elasticity, ψ, for the Japanese labor market as 0.6. This value lies in the plausible range of 0.5-0.7 that Petrongolo and Pissarides (2001) report. I set $\eta = 0.6$ according to the Hosios condition for efficiency.

For the benchmark calibration, I target the average monthly job finding rate of 0.155. This approximately corresponds to the hp-filtered job finding rate in Japan in 1990 as reported by Miyamoto (2011) and Lin and Miyamoto (2012). The Report on Employment Service conducted by the Ministry of Health, Labour and Welfare reports the job openings to application ratios of both permanent and temporary workers, which reflect labor market tightness. Based on the data and the estimates used by Miyamoto (2016), I target the ratio of labor market tightness for temporary to permanent $\theta_{\text{temp}}/\theta_{\text{perm}} = 2.24$. I normalize $\theta_{\text{perm}} = 1$. Each worker has an average working life of 40 years, which implies $\nu = 0.00208$. The maximum and minimum values of human capital $\underline{h}$ and $\overline{h}$ are chosen so that large masses of skilled workers do not accumulate at the endpoints of the skill distribution. I set Δ_h equal to 0.0168, using a grid that has 300 equispaced points.

Workers in my model are born with an initial skill level. I assume that the initial skill is drawn from a log-Normal distribution with mean 0 and variance σ_h^2. I estimate σ_h by first running the following regression using the Japanese Household Panel Survey:

$$\log(\text{monthly income}_{i,t}) = \alpha + Z_{i,t}\beta + \eta_i + \epsilon_{i,t}, \tag{21}$$

where $Z_{i,t}$ includes job-tenure (cubic function), work experience (cubic function), employment type (permanent, temporary), and fixed effects for city-size, year, and industry. Then, σ_h is the estimated standard deviation of η_i in equation 21, which equals 0.494.

Table 3 lists the internally calibrated parameters. I estimate the following parameters using a simulated method of moments: ϕ, b, $\bar{m}_{\text{perm}}$, $\bar{m}_{\text{temp}}$, $\bar{A}$, π_{perm}, π_{temp}, and π_{u}. These parameters are estimated so that the model-produced moments match the data moments below. There are as many parameters as there are targeted moments. I begin by targeting the share of temporary workers to be 9.0 percent, which was the average share of temporary workers for Japanese men during the period from 1990 to 1996. By focusing on the male population, I avoid the effects that arise when there is a change in female participation in the labor force. Following Miyamoto (2016), I calculate the ratio of the job-finding rate of permanent workers to that of temporary workers is 0.450 from the Survey on Employment Trends conducted by the Ministry of Health, Labor and Welfare. I also target the wage distribution of the p90/p50 ratio to be 1.880 and the p50/p10 ratio to be 1.706, which, as estimated by Lise et al. (2014), were the p90/p50 and p50/p10 wage ratios for Japanese males between 1990 and 1995.[15] Using the estimated difference in return to working at the same firm between permanent workers and temporary workers provided in Section 10.2.1, I target the average difference in return on working between the two types of jobs to be 1 percent.[16] Finally,

[15] The p90/p50 and p50/p10 wage ratios for Japanese male were not available for years prior to 1990.

[16] The common wage growth component observed in Section 10.2.1 is captured in γ_X. The only margin that matters in my model is the *difference* between the rate of wage growth between permanent workers and temporary workers.

estimating from the KHPS and the JHPS datasets, I target the average wage ratio between permanent workers and temporary workers to be 1.681. The data moments and the model's outputs are displayed in Table 2.

Table 2: Targeted moments

Moment	Data (Target)	Model Output
Share of temporary workers	9.00%	8.91%
Job finding rate	0.155	0.158
Ratio of job finding rate: perm/temp	0.450	0.445
Unemployment income	0.673 (40% of average wages)	0.674
Wage distribution, p90/p50	1.880	1.852
Wage distribution, p50/p10	1.706	1.706
Average wage ratio: perm/temp	1.681	1.686
Average return on job tenure for perm workers	1.000%	1.036%

Table 3: Internally calibrated parameters

Variable	Description	Value
ϕ	Firing cost	23.569
b	Unemployment income	0.674
$\bar{m}_{\text{perm}}$	Matching efficiency in the permanent sector	0.146
$\bar{m}_{\text{temp}}$	Matching efficiency in the temporary sector	0.237
$\bar{A}$	Productivity of temporary worker	1.053
π_{perm}	Probability of skill upgrade for the perm sector	0.0798
π_{temp}	Probability of skill upgrade for the temp sector	0.0525
π_{u}	Probability of skill deterioration for the unemployed	0.0589

The estimated values of the parameters are shown in Table 3. The firing cost is estimated to be 23.57, which approximately equals 13 months of permanent workers' average wage compensation.[17] Following Esteban-Pretel et al. (2011) and Miyamoto (2016), I target the unemployment income to be about 40 percent of the average monthly wage of all workers, which is 0.673. The estimated value of b is 0.674. The estimated value of the matching efficiency in the permanent and temporary markets implies that the overall job-finding rate is 0.158 and that the ratio of permanent market job finding rates over

[17] Kramarz and Michaud (2010) estimated that in France, the termination of the contract of a permanent job is 16 percent of the annual wage for an individual layoff and 50 percent of the annual wage for a collective layoff. Given that approximately four out of six layoffs are individual layoffs, the average cost is 20 percent of the annual wage. Bentolila et al. (2012) found that firing costs in Spain are approximately 20 percent higher than in France, and Alonso-Borrego et al. (2005) estimated the firing cost is 51 percent of the annual wages in Spain.

the temporary market job-finding rate is 0.445. These numbers are targeted so that the job-finding rate for the overall economy also matches the average monthly job-finding rate in Japan prior to 1990, which was approximately 0.155. The productivity of temporary workers, $\bar{A}$, is estimated to be 1.053. Each month, the probabilities of skill upgrades for permanent jobs and temporary jobs are estimated to be 7.98 percent and 5.25 percent, respectively. If workers are unemployed, their skills deteriorate with the probability of 5.89 percent monthly.

7 Results

To determine how slow growth affects the dual labor market, I set the annual rate of technological growth to 1.68 percent. This is roughly equivalent to the average growth rate of aggregate labor productivity after 1990 in Japan. Table 4 summarizes the results. Column (1) shows the model statistics when the aggregate productivity growth rate and interest rate are set to pre-1990 periods. In column (2), I decrease the productivity growth rate and the interest rate so that they equal the average rates of productivity growth and real interest in Japan between 1991 and the early 2010s. Adjusting the real interest rate in my model to the observed real interest rates in Japan after 1990 is important because as the interest rate decreases, the future incomes are discounted at a lower rate, potentially nullifying the effect of slower aggregate productivity growth. Additionally, I conduct another experiment in column (3) where the real interest rate is assumed to stay at the pre-1990 level of 4.85 percent, to isolate the effect of the slowdown in productivity growth rate on the temporary job shares.

Table 4: Model statistics

	(1)	(2)	(3)
Share of temporary worker	8.91%	13.30%	20.51%
Job finding rate	0.158	0.158	0.160
Job separation rate	0.0038	0.0040	0.0042
Average wage ratio: perm/temporary	1.681	1.729	1.803
Unemployment rate	3.61%	3.70%	3.80%
Job-finding rate in perm. sector	0.146	0.141	0.134
Job-finding rate in temp. sector	0.328	0.310	0.296
Annual rate of technological growth	4.13%	1.68%	1.68%
Annual rate of interest	4.85%	3.12%	4.85%

The results suggest that the effects of the slowdown in aggregate productivity growth on the share of temporary workers are profound. A comparison of column (1) and column (2) in Table 4 shows that the share of temporary workers increases from 8.91 percent to 13.30 percent when the aggregate productivity growth slows down despite the reduction in the interest rate. Given that the average share of temporary workers for Japanese males increased from 9 percent between 1990 and 1996 to 20.42 percent between 2010 and 2015, the model suggests that approximately 38 percent of the increase in the share of temporary workers among Japanese males can be explained by the slow growth. Column (3) shows that the share of temporary workers increases to 20.51 percent under the slow-growth economy when the interest rate is forced to remain unchanged. This suggests that if the Japanese real interest rate stayed at the same rate as that in the pre-1990 periods, the slowdown in aggregate productivity growth alone could have increased the Japanese male share of temporary workers to 20.51 percent.

The market tightness in both sectors decreases when the rate of technological growth slows down, which indicates that firms post fewer vacancies per job-seeker when the economy is in the slow growth steady state. The decrease in market tightness results in a low job-finding rate in both permanent and temporary markets. For the economy as a whole, however, the job-finding rate increases slightly. This increase reflects the changes in the composition of the labor market. Although the job-finding rate within each market declines, the large increase in the share of temporary workers whose job-finding rate is relatively high offsets the declines in job-finding rates within permanent and temporary markets. Similarly, the separation rate increases for the labor market as a whole because a larger fraction of workers are now employed in the temporary sector whose job separation rate is relatively high.

Finally, the unemployment rate of the aggregate economy increases in the slow growth steady state. This is consistent with the Japanese experience, wherein the unemployment rate has increased during the last two decades. In this model, an increase in unemployment is modest because, under the slow-growth economy, a greater number of workers search in a temporary market whose matching efficiency is higher.

Figure 2 shows the changes in the skill distribution of workers based employment types using the results shown in columns (1) and (2) in Table 4. The horizon axis represents the workers' skills on a scale of 0 to 5. The blue solid line represents the distribution of temporary workers, the red solid line represents the distribution of permanent workers, and the black solid line represents the distribution of unemployed workers when the aggregate productivity growth rate is 4.13 percent (corresponding to column (1) in Table 4). Similarly, the blue dotted line represents the distribution of temporary workers, the red dotted line represents the distribution of permanent workers, and the black dotted line represents the distribution of unemployed workers when the productivity growth rate and real interest rate both fall to post 1990 rates (corresponding to column (2) in Table 4). The figure demonstrates that in the slower growth steady-state, the right tail of the skill distribution for temporary workers and the left tail of the distribution for permanent workers both shift to the right. This indicates that some workers in the middle who would prefer searching

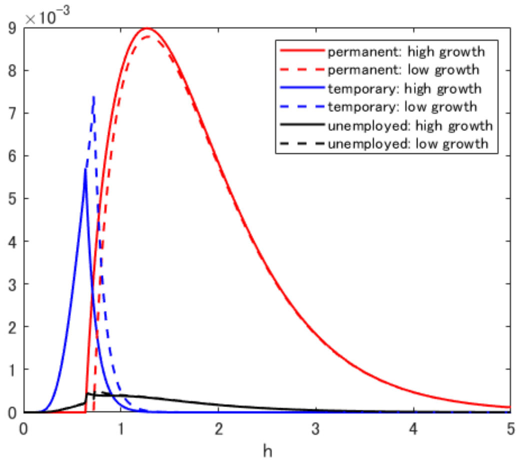

Fig. 2: Skill distribution by employment types

Notes: The figure shows the changes in skill distribution of workers by employment types using the results shown in columns (1) and (2) in Table 4. The horizon axis represents the workers' skills (h) on a scale of 0 to 5. The blue solid line represents the distribution for temporary workers, the red solid line represents the distribution for permanent workers, and the black solid line represents the distribution for unemployed workers when the aggregate productivity growth rate is 4.13 percent. The blue dotted line represents the distribution of temporary workers, the red dotted line represents the distribution of permanent workers, and the black dotted line represents the distribution of unemployed workers when the aggregate productivity growth rate is 1.68 percent.

jobs in the permanent market in the fast-growing economy switch to search jobs in the temporary market in the slow-growth economy.

8 Policy experiments

In this section, I conduct two policy experiments to investigate the effect of other possible candidates that could increase the share of temporary workers. First, I examine the effect of changing the firing cost of permanent workers. Second, I explore the effect of a change in the productivity of temporary workers.

8.1 Firing cost

Research indicates that strong protection against laying off permanent workers induces firms to substitute permanent workers for temporary workers.[18] The underlying intuition is that layoff protection increases the burden placed

[18] See Cahuc et al. (2016).

on employers when they hire permanent workers, which motivates firms to post fewer vacancies in the permanent market. Therefore, more job-seeker are attracted to the permanent market, and the share of temporary workers decreases. In this section, I discuss whether the same intuition holds in my model and investigate its quantitative importance.

In this model, the parameter, ϕ, controls the difficulty of firing a permanent worker. In the experiment presented below, I simulate the share of temporary workers using different values of ϕ. Other parameter values are set according to the baseline calibration presented in Section 7. Throughout the experiment, I assume that the economy is in the slow growth state (after 1990) where the annual productivity growth rate is 1.68 percent and the real interest rate is 3.12 percent (corresponds to column (2) in Table 4). Table 5 summarizes the results.

Table 5: Model statistics with different layoff policies

	(1)	(2)	(3)	(4)	(5)
Value of firing cost (ϕ)	0	11.79	23.57 **(baseline)**	47.14	117.85
Firing cost in terms of monthly permanent wages	0 mth	6.5 mths	13.5 mths	27 mths	67.5 mths
Share of temporary workers	8.14%	9.74%	13.30%	16.24%	33.02%

Notes: The baseline result, shown in column (3) of this table, assumes that the aggregate productivity growth rate and the real interest rate are both at the pre-1990 level. The baseline result corresponds to the result shown in column (2) in Table 4.

Each column in Table 5 shows the simulated share of temporary workers with corresponding values of ϕ. The results are consistent with the basic intuition described above in that as the firing costs of permanent workers increase, the share of temporary workers increases. When the layoff cost (ϕ) increases to 47.14 (approximately 27 months of average wages for permanent workers) and to 117.85 (approximately 67.5 months of average wages for permanent workers), the share of temporary workers increases to 16.24 percent and to 33.02 percent, respectively. Table 5 also shows that the reduction of firing cost to 11.79 (6.5 months of average wages for permanent workers) and the elimination of firing cost reduce the share of temporary workers to 9.74 percent and to 8.14 percent, respectively. The result implies that the elimination of firing costs would have the effect of reversing the increase in temporary workers' share caused by slow productivity growth since 1990.

8.2 Temporary workers' technology

I further explore the effect of changing temporary job technology on the share of temporary workers. Note that the temporary technology in my model represents the productivity of temporary workers *relative* to that of permanent workers.

Table 6: Model statistics with different temporary technology

	(1)	(2)	(3)	(4)	(5)
Temporary job technology ($\bar{A}$)	0.842	0.948	1.053 **(baseline)**	1.158	1.264
$\bar{A}$ ratio compared to baseline: $(\bar{A}/\bar{A}_{\text{baseline}})$	0.8	0.9	1	1.1	1.2
Share of temporary workers	3.26%	6.71%	13.30%	18.27%	24.83%

Notes: The baseline result, shown in column (3) of this table, assumes that the aggregate productivity growth rate and the real interest rate are both at the pre-1990 level. The baseline result corresponds to the result shown in column (2) in Table 4.

Table 6 shows the simulated share of temporary workers with corresponding values of the temporary technology. Similar to the firing cost experiment, I conduct policy experiments with temporary technology assuming that the economy is in the slow-growth steady states, setting the aggregate productivity growth and real interest rates at the post 1990 rates. The results show a positive relationship between the temporary technology and the temporary workers' share. Column (3) re-posts the baseline result from column (2) of Table 4. As the temporary technology improves by 10 percent and by 20 percent from the baseline level, the share of temporary workers increases to 18.27 percent and to 24.83 percent, respectively. Looking in the opposite direction, a reduction of temporary technology by 10 percent and by 20 percent would reduce the share of temporary workers to 6.71 percent and to 3.06 percent, respectively. The positive association between the temporary technology and share of a temporary worker is driven by the fact that the temporary job's technological improvement makes temporary jobs more attractive, which drive unemployed workers to seek temporary jobs and firms to post more vacancies in the temporary labor market. An implication from this exercise is that an advancement in technologies that helps workers perform routine tasks effectively would intensify the increasing trend of temporary workers' share in future.

9 Concluding Remarks

This paper investigates how the rate of economic growth affects the composition of the labor market. I construct a search and matching model of a dual labor market that is inhabited by heterogeneous workers. I find that the rate of workers' skill accumulation on the job significantly affects how the slow growth of aggregate labor productivity affects labor market composition. My model predicts that approximately 38 percent of the observed increase in the share of temporary workers among Japanese males can be explained by the slow growth of aggregate productivity. When different rates of skill accumulation among different sectors of the labor market are combined with slower aggregate productivity growth, the labor market composition can change significantly.

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10 Appendix

10.1 Data Description

Throughout the analysis, I used the Japanese Household Panel Survey (JHPS) and the Keio Household Panel Survey (KHPS). The original datasets include approximately 4000 households between 2004 to 2008 and 8000 households between 2009 and 2014. I focus on the income of the sub-sample of the household heads who were recorded at least 4 times. This criterion leaves us with about 2000 samples each year from 2004 to 2008 and about 3200 samples from 2009 to 2013.

The survey records the labor income of household heads under the following four categories: (i) monthly wages, (ii) weekly wages, (iii) hourly wages, and (iv) annual bonuses. I construct estimated hourly wages based on all four measures of the wages.

The monthly wages and annual bonuses are converted into hourly wages by the following procedure:

$$\text{hourly wages} = \frac{\text{monthly wages} + \text{annual bonuses} \ / \ 12}{(\text{average days worked per month}/4.381 \text{ weeks}) \times \text{hours of work per week}}$$

The denominator is the hours worked per month, which is estimated using the reported average days worked per month, the average number of weeks in a month (4.381 weeks), and the reported hours of work per week.

The weekly wages are converted into hourly wages by the following procedure:

$$\text{hourly wages} = \frac{\text{weekly wages}}{\text{hours of work per week}}$$

A part of the job tenure is also estimated for the analysis. The survey asks respondents the year in which they started their current employment. From this figure, I calculate the job tenure. However, this question is available only when the respondents enter the survey for the first time. For the following years, I add a year to the job tenure if the respondents reported that they stayed at the same workplace from the previous year. Otherwise, I set the job tenure to 0.

The work experience is constructed in a similar manner. The number of years working is reported only when the respondents are recorded in the survey for the first time. From this, I add a year to the work experience if the respondents reported that they worked in the following years. Otherwise, the work experience in each survey year remains the same as the previous year.

The following empirical findings are based largely on two datasets: the Japanese Household Panel Survey (JHPS) and the Keio Household Panel Survey (KHPS). The KHPS is an annual panel household survey of 4000 households that were recorded each year from 2004 to 2013. The JHPS is another annual panel household survey of about 4000 households that were recorded each year from 2009 to 2013. I focus on the income of household heads aged 25 to 60 years who were recorded at least 4 times (waves). This criterion leaves us with about 2000 observations each year from 2004 to 2008 and about 3200 observations from 2009 to 2013. The classification of permanent worker and temporary worker follows the criterion described above.

In these datasets, I define "temporary employees" as workers who fall into one of the following categories: (i) part-time worker, (ii) dispatched workers, (iii) contract workers, (iv) commission staff, or (v) other types of workers who are not permanent employees. This method of classification is consistent with the use of the term in the Labor Force Survey [19] and captures the different employment practices for permanent and temporary workers described above. Under this system of classification, some temporary workers work for more than 40 hours a week and some permanent workers can work less than 40 hours a week.

10.2 Permanent and temporary job characteristics

In this subsection, I document three empirical facts about permanent and temporary workers: (i) the rate of wage growth is higher for permanent workers, (ii) people with high school or middle school degrees are more likely to become temporary workers while people with college or higher degrees are more likely to become permanent workers, and (iii) the opportunity to receive training on the job for temporary workers is about half of that for permanent workers.

10.2.1 Wage growth and job tenure

First, I estimate the rate of wage growth for permanent workers and temporary workers. To estimate the return on working for staying an additional year at the same firm, I run regressions of log hourly wage on a quadratic function of job tenure for permanent workers and temporary workers, separately. The regressions also include the following control variables in $\mathbf{X_{i,t}}$: a quadratic function of age, and fixed effects for the year, industry, and region. The regression is specified as follows:

$$\ln(\text{wage}_{i,t}) = \alpha_i^s + \beta_1^s \text{tenure}_{i,t} + \beta_2^s \text{tenure}_{i,t}^2 + \mathbf{X_{i,t}} \boldsymbol{\Phi}^{\mathbf{s}} + \epsilon_{i,t}, \quad \text{where } s = \{\text{perm, temp}\} \tag{22}$$

where α_i^s represents individual fixed effects, and the s superscript is an indicator for permanent or temporary workers. Our parameters of interest are β_1^s and β_2^s, which indicate the returns to job tenure for permanent and temporary jobs. The result is shown in Table 7 below. The return to tenure is statistically significant for permanent jobs. However, it is statistically insignificant for temporary jobs. For example, the average return of staying at the current firm for permanent workers with 3 years of job tenure is approximately 1.1%. In contrast, the annual return of working at the current firm is essentially zero for temporary workers with 3 years of job tenure. The result is consistent with

[19] The survey design is similar to the U.S. Current Population Survey

common practices documented in Section 3 wherein permanent workers accumulate local knowledge within a firm and receive extensive on-the-job training; temporary workers, in contract, do not.

Table 7: Return on working

Dependent variable: ln(wage)		
	Permanent	Temporary
Tenure	0.0124	-0.0006
	(0.00375)	(0.0092)
Tenure^2	-0.00018	-0.0001
	(0.00012)	(0.0003)
Age, age-squared	yes	yes
Year, Industry, Region FE	yes	yes
Individual FE	yes	yes
Observations	14,089	2,379

Notes: Data are from the KHPS and JHPS, 2004-2013. Numbers in parenthesis report the robust standard errors.

10.2.2 Educational attainment

The assumptions in my model induce low-skilled workers to search for temporary jobs and high-skilled workers to search for permanent jobs; I present an observation that is consistent with this prediction. Since workers' innate productive capacities are difficult to measure, I use workers' educational attainment as a proxy for their skills. According to the Basic Surveys on Employment Structure, the share of temporary workers is 26 percent in 2012 among male employees with middle or high school degrees. In contrast, the share of temporary workers is 12 percent in 2012 among male employees who have college or higher degrees. While this is not a perfect measure of workers' skills, it hints at the difference in the workers' traits across the two types of employment as measured by educational achievement.

10.2.3 On the job training

Lastly, I document that firms are about twice more likely to provide job training for permanent workers than they do for temporary workers. According to the bi-annual Basic Survey of Human Resources Development in Japan between 2008 and 2016, on average, about 60 percent of establishments provide "on-the-job training" to their permanent employees within a given year while only about 25 percent of them provides temporary employees with the "on-the-job training." Additionally, on average, approximately 70 percent of the establishments provide permanent workers with "off-the-job training" while only about 32 percent of them provide temporary workers with the training.

10.3 Stationary Inducing Transformation

This section shows the derivation of the stationary inducing transformation and that the value functions are homogeneous of degree one in X. Let variables in *tilde* indicate the variables before the stationary-inducing transformation. Without loss of generality, I assume no human capital accumulation, $\pi_j = 0$ for $j \in \{\text{perm, temp, u}\}$. The system of value functions before conducting the stationary-inducing transformation is:

$$\tilde{U}(h) = \tilde{b} + \beta(1-\nu)\max \mathbb{E}\left[\tilde{U}'_{\text{perm}}(h), \tilde{U}'_{\text{temp}}(h)\right], \tag{23}$$

$$\tilde{U}_{\text{perm}}(h) = f(\theta_{\text{perm}})(\tilde{W}_{\text{perm}}(h) - \tilde{U}(h)) + \tilde{U}(h), \tag{24}$$

$$\tilde{U}_{\text{temp}}(h) = f(\theta_{\text{temp}})(\tilde{W}_{\text{temp}}(h) - \tilde{U}(h)) + \tilde{U}(h), \tag{25}$$

$$\tilde{W}^N_{\text{perm}}(h) = \tilde{w}^N_{\text{perm}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{perm}})(\tilde{W}^{E'}_{\text{perm}}(h) - \tilde{U}'(h)) + \tilde{U}'(h)\right], \tag{26}$$

$$\tilde{W}^E_{\text{perm}}(h) = \tilde{w}^E_{\text{perm}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{perm}})(\tilde{W}^{E'}_{\text{perm}}(h) - \tilde{U}'(h)) + \tilde{U}'(h)\right], \tag{27}$$

$$\tilde{W}_{\text{temp}}(h) = \tilde{w}_{\text{temp}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{temp}})(\tilde{W}'_{\text{temp}}(h) - \tilde{U}(h)) + \tilde{U}(h)\right], \tag{28}$$

$$\tilde{J}^N_{\text{perm}}(h) = h \times X - \tilde{w}^N_{\text{perm}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{perm}})\tilde{J}^{E'}_{\text{perm}}(h) - \delta_{\text{perm}}\tilde{\phi}'\right], \tag{29}$$

$$\tilde{J}^E_{\text{perm}}(h) = h \times X - \tilde{w}^E_{\text{perm}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{perm}})\tilde{J}^{E'}_{\text{perm}}(h) - \delta_{\text{perm}}\tilde{\phi}'\right], \tag{30}$$

$$\tilde{J}_{\text{temp}}(h) = A \times X - \tilde{w}_{\text{temp}}(h) + \beta(1-\nu)\,\mathbb{E}\left[(1-\delta_{\text{temp}})\tilde{J}'_{\text{perm}}(h)\right], \tag{31}$$

$$\kappa_{\tilde{\text{perm}}} = \beta(1-\nu)\,\mathbb{E}\left[q(\theta_{\text{perm}})\int_h \tilde{J}'_{\text{perm}}(h)d\mu^h_{\text{perm}}\right], \tag{32}$$

$$\kappa_{\tilde{\text{temp}}} = \beta(1-\nu)\,\mathbb{E}\left[q(\theta_{\text{temp}})\int_h \tilde{J}'_{\text{temp}}(h)d\mu^h_{\text{temp}}\right], \tag{33}$$

$$\begin{aligned}\tilde{w}^N_{\text{perm}} =& (1-\eta)\tilde{b} + \eta\left(\tilde{y}_{\text{perm}} - \beta(1-\nu)\tilde{\phi}'\right)\\ &+ (1-\eta)\beta(1-\nu)\left\{(\pi_{\text{perm}} - \pi_u)\tilde{U}'(h) + \pi_u\tilde{U}'(h-\Delta_h) - \pi_{\text{perm}}\tilde{U}'(h+\Delta_h)\right\}\\ &+ (1-\eta)\beta(1-\nu)\left\{(1-\pi_u)f(\theta_{\text{perm}})(\tilde{W}^{N'}_{\text{perm}}(h) - \tilde{U}'(h))\right.\\ &\qquad\left. + \pi_u f(\theta_{\text{j}})(\tilde{W}^{N'}_{\text{j}}(h-\Delta_h) - \tilde{U}'(h-\Delta_h))\right\},\end{aligned} \tag{34}$$

$$\begin{aligned}\tilde{w}^E_{\text{perm}} =& (1-\eta)\tilde{b} + \eta\tilde{y}_{\text{perm}} + \eta(1-\beta(1-\nu))\tilde{\phi}'\\ &+ (1-\eta)\beta(1-\nu)\left\{(\pi_{\text{perm}} - \pi_u)\tilde{U}'(h) + \pi_u\tilde{U}'(h-\Delta_h) - \pi_{\text{perm}}\tilde{U}'(h+\Delta_h)\right\}\\ &+ (1-\eta)\beta(1-\nu)\left\{(1-\pi_u)f(\theta_{\text{perm}})\left(\tilde{W}^{N'}_{\text{perm}}(h) - \tilde{U}'(h)\right)\right.\\ &\qquad\left. + \pi_u f(\theta_{\text{j}})\left(\tilde{W}^{N'}_{\text{j}}(h-\Delta_h) - \tilde{U}'(h-\Delta_h)\right)\right\},\end{aligned} \tag{35}$$

$$\tilde{w}_{\text{temp}} = (1-\eta)\tilde{b} + \eta\tilde{y}_{\text{temp}} + (1-\eta)\beta(1-\nu)\left\{f(\theta_{\text{temp}})(\tilde{W}'_{\text{temp}}(h) - \tilde{U}'(h))\right\}, \tag{36}$$

To derive a system of equations that are on the balanced growth, I divide all variables by X. I assume that X exhibits a steady-state rate of growth denoted by γ_X, $\gamma_X = X'/X$. Furthermore, I also assume that all exogenous variables grow at the constant rate of γ_X: $b = \tilde{b}/X$, $\phi = \tilde{\phi}/X$, and $\kappa_j = \tilde{\kappa_j}/X$. Under these assumptions, the system of equations (23)-(36) can be written as follows:

$$
\begin{aligned}
U(h) =& b+\beta(1-\nu)\gamma_X \max \mathbb{E}\left[U'_{\text{perm}}(h), U'_{\text{temp}}(h)\right], && (37)\\
U_{\text{perm}}(h) =& f(\theta_{\text{perm}})(W_{\text{perm}}(h)-U(h))+U(h), && (38)\\
U_{\text{temp}}(h) =& f(\theta_{\text{temp}})(W_{\text{temp}}(h)-U(h))+U(h), && (39)\\
W^N_{\text{perm}}(h) =& w^N_{\text{perm}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{perm}})(W^{E'}_{\text{perm}}(h)-U'(h))+U'(h)\Big], && (40)\\
W^E_{\text{perm}}(h) =& w^E_{\text{perm}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{perm}})(W^{E'}_{\text{perm}}(h)-U'(h))+U'(h)\Big], && (41)\\
W_{\text{temp}}(h) =& w_{\text{temp}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{temp}})(W'_{\text{temp}}(h)-U'(h))+U'(h)\Big], && (42)\\
J^N_{\text{perm}}(h) =& h-w^N_{\text{perm}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{perm}})J'_{\text{perm}}(h)-\delta_{\text{perm}}\phi'\Big], && (43)\\
J^E_{\text{perm}}(h) =& h-w^E_{\text{perm}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{perm}})J'_{\text{perm}}(h)-\delta_{\text{perm}}\phi'\Big], && (44)\\
J_{\text{temp}}(h) =& A-w_{\text{temp}}(h)+\beta(1-\nu)\gamma_X\,\mathbb{E}\Big[(1-\delta_{\text{temp}})J'_{\text{perm}}(h)\Big], && (45)\\
\kappa_{\text{perm}} =& \beta(1-\nu)\gamma_X\,\mathbb{E}\Big[q(\theta_{\text{perm}})\int_h J'_{\text{perm}}(h)d\mu^h_{\text{perm}}\Big], && (46)\\
\kappa_{\text{temp}} =& \beta(1-\nu)\gamma_X\,\mathbb{E}\Big[q(\theta_{\text{temp}})\int_h J'_{\text{temp}}(h)d\mu^h_{\text{temp}}\Big], && (47)\\
w^N_{\text{perm}} =& (1-\eta)b+\eta\left(y_{\text{perm}}-\beta(1-\nu)\gamma_X\phi\right)\\
&+(1-\eta)\beta(1-\nu)\gamma_X\Bigg\{(\pi_{\text{perm}}-\pi_u)U(h)+\pi_u U(h-\varDelta_h)-\pi_{\text{perm}}U(h+\varDelta_h)\Bigg\}\\
&+(1-\eta)\beta(1-\nu)\gamma_X\Bigg\{(1-\pi_u)f(\theta_{\text{perm}})(W^N_{\text{perm}}(h)-U(h))\\
&\qquad+\pi_u f(\theta_{\text{j}})(W^N_{\text{j}}(h-\varDelta_h)-U(h-\varDelta_h))\Bigg\}, && (48)\\
w^E_{\text{perm}} =& (1-\eta)b+\eta y_{\text{perm}}+\eta(1-\beta(1-\nu)\gamma_X)\phi\\
&+(1-\eta)\beta(1-\nu)\gamma_X\Bigg\{(\pi_{\text{perm}}-\pi_u)U(h)+\pi_u U(h-\varDelta_h)-\pi_{\text{perm}}U(h+\varDelta_h)\Bigg\}\\
&+(1-\eta)\beta(1-\nu)\gamma_X\Bigg\{(1-\pi_u)f(\theta_{\text{perm}})\Big(W^N_{\text{perm}}(h)-U(h)\Big)\\
&\qquad+\pi_u f(\theta_{\text{j}})\Big(W^N_{\text{j}}(h-\varDelta_h)-U(h-\varDelta_h)\Big)\Bigg\}, && (49)\\
w_{\text{temp}} =& (1-\eta)b+\eta y_{\text{temp}}+(1-\eta)\beta(1-\nu)\gamma_X\Bigg\{f(\theta_{\text{temp}})(W'_{\text{temp}}(h)-U'(h))\Bigg\}, && (50)
\end{aligned}
$$

Thus, the system of equations are are homogeneous of degree one in X.

10.4 Wage Determination

As explained in Section 5.9, a wage is negotiated between a worker and a firm according to the Nash Bargaining. The two parties split the match surplus as follows:

$$(1-\eta)(W^N_{\text{perm}} - U_{\text{perm}}) = \eta J^N_{\text{perm}}, \tag{51}$$

for a new permanent worker,

$$(1-\eta)(W^E_{\text{perm}} - U_{\text{perm}}) = \eta (J^E_{\text{perm}} + \phi), \tag{52}$$

for an existing permanent worker, and

$$(1-\eta)(W_{\text{temp}} - U_{\text{temp}}) = \eta J_{\text{temp}}, \tag{53}$$

for a temporary worker.

I derive the wage for a new permanent worker here. To derive the wage equation, I first write out the value functions in the equation 51 using the value functions defined in sections 5.5, 5.6, and 5.7:

$$\begin{aligned}
&(1-\eta)\Bigg[w^N_{\text{perm}} + \beta(1-\nu)\gamma_X\Bigg\{(1-\pi_{\text{perm}})\Bigg((1-\delta_{\text{perm}})(W^E_{\text{perm}}(h) - U(h)) + U(h)\Bigg) \\
&\qquad + \pi_{\text{perm}}\Bigg((1-\delta_{\text{perm}})(W^E_{\text{perm}}(h+\Delta_h) - U(h+\Delta_h)) + U(h+\Delta_h)\Bigg)\Bigg\} \\
&\qquad - b - \beta(1-\nu)\gamma_X\Bigg\{(1-\pi_u)\Bigg(f(\theta_{\text{perm}})(W^N_{\text{perm}}(h) - U(h)) + U(h)\Bigg) \\
&\qquad + \pi_u\Bigg(f(\theta_{\text{perm}})(W^N_{\text{perm}}(h-\Delta_h) - U(h-\Delta_h)) + U(h-\Delta_h)\Bigg)\Bigg\}\Bigg] \\
&= \eta\Bigg[y^N_{\text{perm}} - w^N_{\text{perm}} + \beta(1-\nu)\gamma_X\Bigg\{(1-\pi_{\text{perm}})\Bigg((1-\delta_{\text{perm}})J^E_{\text{perm}}(h) - \delta_{\text{perm}}\phi\Bigg) \\
&\qquad + \pi_{\text{perm}}\Bigg((1-\delta_{\text{perm}})J^E_{\text{perm}}(h+\Delta_h) - \delta_{\text{perm}}\phi\Bigg)\Bigg\}\Bigg],
\end{aligned} \tag{54}$$

Using the expression in equation 52, I simplify the expression above as follows:

$$
\begin{aligned}
&w^N_{\text{perm}} - (1-\eta)b - \eta y^N_{\text{perm}} + \eta\beta(1-\nu)\gamma_X\delta_{\text{perm}}\phi \\
&+ \beta(1-\nu)\gamma_X\Bigg\{(1-\pi_{\text{perm}})(1-\delta_{\text{perm}})\left(\underbrace{(1-\eta)\left(W^E_{\text{perm}}(h)-U(h)\right)-\eta J^E_{\text{perm}}(h)}_{\eta\phi}\right) \\
&+ \pi_{\text{perm}}(1-\delta_{\text{perm}})\left(\underbrace{(1-\eta)\left(W^E_{\text{perm}}(h+\Delta_h)-U(h+\Delta_h)\right)-\eta J^E_{\text{perm}}(h+\Delta_h)}_{\eta\phi}\right) \\
&+ (1-\eta)\Big((1-\pi_{\text{perm}})U(h)+\pi_{\text{perm}}U(h+\Delta_h)-(1-\pi_u)U(h)-\pi_u U(h-\Delta_h) \\
&- (1-\pi_u)f(\theta_{\text{perm}})\big(W^N_{\text{perm}}(h)-U(h)\big)-\pi_u f(\theta_{\text{j}})\big(W^N_{\text{j}}(h-\Delta_h)-U(h-\Delta_h)\big)\Big)\Bigg\} = 0,
\end{aligned}
\tag{55}
$$

Finally, the wage equation for a new permanent worker can be written as follows:

$$
\begin{aligned}
w^N_{\text{perm}} &= (1-\eta)b + \eta\left(y_{\text{perm}} - \beta(1-\nu)\gamma_X\phi\right) \\
&+ (1-\eta)\beta(1-\nu)\gamma_X\Big\{(\pi_{\text{perm}}-\pi_u)U(h)+\pi_u U(h-\Delta_h)-\pi_{\text{perm}}U(h+\Delta_h)\Big\} \\
&+ (1-\eta)\beta(1-\nu)\gamma_X\Big\{(1-\pi_u)f(\theta_{\text{perm}})(W^N_{\text{perm}}(h)-U(h)) \\
&\qquad + \pi_u f(\theta_{\text{j}})(W^N_{\text{j}}(h-\Delta_h)-U(h-\Delta_h))\Big\},
\end{aligned}
\tag{56}
$$

One can follow the same steps to derive the wage equation for an existing permanent worker:

$$
\begin{aligned}
w^E_{\text{perm}} &= (1-\eta)b + \eta y_{\text{perm}} + \eta(1-\beta(1-\nu)\gamma_X)\phi \\
&+ (1-\eta)\beta(1-\nu)\gamma_X\Big\{(\pi_{\text{perm}}-\pi_u)U(h)+\pi_u U(h-\Delta_h)-\pi_{\text{perm}}U(h+\Delta_h)\Big\} \\
&+ (1-\eta)\beta(1-\nu)\gamma_X\Big\{(1-\pi_u)f(\theta_{\text{perm}})\Big(W^N_{\text{perm}}(h)-U(h)\Big) \\
&\qquad + \pi_u f(\theta_{\text{j}})\Big(W^N_{\text{j}}(h-\Delta_h)-U(h-\Delta_h)\Big)\Big\},
\end{aligned}
\tag{57}
$$

Likewise, the wage equation for a temporary worker can be derived as follows:

$$
\begin{aligned}
w_{\text{temp}} = {} & (1-\eta)b + \eta y_{\text{temp}} \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Big\{ (\pi_{\text{temp}} - \pi_u)U(h) + \pi_u U(h-\Delta_h) - \pi_{\text{temp}} U(h+\Delta_h) \Big\} \\
& + (1-\eta)\beta(1-\nu)\gamma_X \Big\{ (1-\pi_u) f(\theta_{\text{temp}}) \Big(W_{\text{temp}}(h) - U(h) \Big) \\
& \qquad + \pi_u f(\theta_{\text{temp}}) \Big(W_{\text{temp}}(h-\Delta_h) - U(h-\Delta_h) \Big) \Big\},
\end{aligned} \tag{58}
$$

10.5 Computation

The computation algorithm for the model with exogenous separation is explained in this section. There is only one state variable, which is workers' skill-level, h. Because workers are heterogeneous only in their skills, workers with the same skill all search in the same market in a steady state.

The computation algorithm is explained as follows:

1. Guess, for all h, the market in which the workers with skill level h search for a job
2. Guess $U(h)$ for each skill level, h.
3. Solve for J and w using Equations 9, 10, 11, 17, 18, and 19.
4. Solve for θ and the steady-state skill distributions of workers using Equations 12, 13, and 20.
5. Solve for W using Equations 6, 7, and 8.
6. Verify that guess of $U(h)$ satisfies Equations 4 and 5 for all skill levels, h,.
   - If not, update the guess, and return to step 3. Otherwise, proceed to step 7.
7. Verify that initial guess of market that workers with skill, h, search satisfies Equation 3
   - If not, update the guess, and return to step 1. Repeat until convergence.

10.6 No Endogenous Skill Accumulation

To highlight the importance of endogenous skill accumulation in my model, I impose zero human capital accumulation (i.e. $\pi_{\text{perm}} = \pi_{\text{temp}} = \pi_{\text{u}} = 0$) and lower the steady state growth rate of aggregate productivity from 4.13 percent to 1.68 percent. In this environment workers are born with randomly realized skills, which do not change for the rest of their careers.

Table 8 summarizes the target moments for calibrating the model in an economy that prohibits endogenous skill accumulation. The task of the procedure is to find values of the five parameters, ϕ, b, $\bar{m}_{\text{perm}}$, $\bar{m}_{\text{temp}}$, and $\bar{A}$ to match five moments. The estimated parameter values are listed in Table 9.

Table 8: Targeted and model statistics

Moment	Data (Target)	Model Output
Share of temporary workers	9.00%	8.72%
Job finding rate	0.155	0.158
Ratio of job finding rate: perm/temp	0.450	0.445
Unemployment income	0.445	0.452
Average wage ratio: perm/temp	1.681	2.286

Table 9: Internally calibrated parameters

Variable	Description	Value
ϕ	Firing cost	1.271
b	Unemployment income	0.452
$\bar{m}_{\text{perm}}$	Matching efficiency in permanent sector	0.146
$\bar{m}_{\text{temp}}$	Matching efficiency in temporary sector	0.237
$\bar{A}$	Productivity of temporary worker	0.513

Table 10 summarizes the results of this experiment. The most striking result is that the share of temporary workers does not increase. Figure 3 shows that the shape of the workers' skill distribution is unchanged even after the aggregate productivity growth rate has decreased to 1.68 percent. This illustrates that when human skills are not allowed to evolve, the value of unemployment in the permanent market and the value of unemployment in the temporary market both decrease by the same margin, which leaves the cutoff level h^* unaltered. Therefore, workers will continue to compare the value of searching in the permanent market against the value of searching in the temporary market in the manner they did before the change.

In this environment, the job-finding, job separation, and unemployment rates are not affected by the slowdown in aggregate productivity growth. This section confirms that endogenous skill accumulation plays a key role in the relationship between slow growth and labor market composition. The existence of workers with heterogeneous skills alone does not generate the negative relationship between aggregate productivity growth rate and the share of temporary workers examined in the baseline model.

Table 10: Model statistics:
Case of no skill accumulation

	(1)	(2)
Share of temporary worker	8.72%	8.72%
Job finding rate	0.158	0.158
Job separation rate	0.0038	0.0038
Average wage ratio: perm/temporary	2.286	2.283
Unemployment rate	3.60%	3.61%
Job-finding rate in perm. sector	0.146	0.146
Job-finding rate in temp. sector	0.328	0.327
Annual rate of technological growth	4.13%	1.68%
Annual rate of interest	4.85%	3.12%

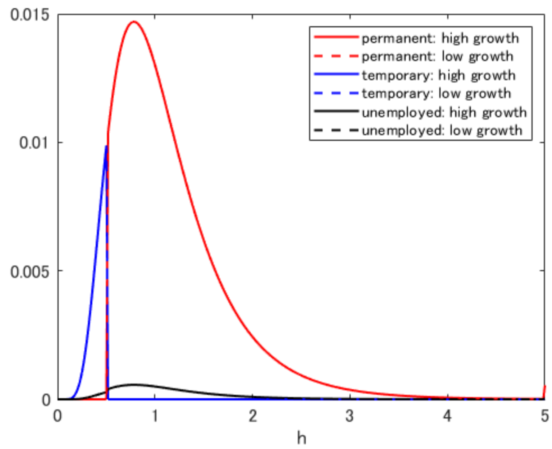

Fig. 3: Skill distribution by employment types:
Case of no skill accumulation

Notes: The figure shows the changes in skill distribution of workers by employment types using the results shown in columns (1) and (2) in Table 4. The horizon axis represents the workers' skills (h) on a scale of 0 to 5. The blue solid line represents the distribution of temporary workers, the red solid line represents the distribution of permanent workers, and the black solid line represents the distribution of unemployed workers when the aggregate productivity growth rate is 4.13 percent. The blue dotted line represents the distribution of temporary workers, the red dotted line represents the distribution of permanent workers, and the black dotted line represents the distribution of unemployed workers when the aggregate productivity growth rate is 1.68 percent.